

Austin Huang

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SUMMARY

Robotics and automation systems engineer focused on production-grade automation across robotics, vision, software integration, and AI-agent operations.

SKILLS

Core Strengths: Hardware–Software Integration; Reliability Engineering; System Architecture; High-Throughput Workflow Design; Data-Driven Optimization; Cross-Functional Execution; Customer-Facing Deployment; Technical Documentation.

Technical Skills: Robotics Integration (ABB Robots, AGV/AMR Systems); ROS/ROS2; Vision Systems (OpenCV, YOLO); Motion Planning; API Integration; Python; C++; MATLAB; Mechanical Design (Creo); System Calibration; Linux (Ubuntu); Git; LLM Agents & Agentic Workflows; MuJoCo.

EXPERIENCE

XtalPi Inc., Boston, MA

Automation Engineer → Manager, Automation Systems & Delivery (*North America*)

August 2024 – Present

- Owned end-to-end deployment and stabilization of a production-grade automated chemistry R&D platform, defining system architecture, integration logic, and deployment readiness for U.S. commercial rollout.
- Served as the primary on-site system owner at a Fortune-500 pharma partner, recovering system performance and improving task evaluation pass rates from 59% to 100% and target hit rates from 29% to 98%.
- Developed a monocular contour-based calibration method for end-effector pose correction, improving alignment accuracy while reducing hardware dependency and deployment complexity for solid-dispensing workflows.
- Led system-level reliability improvements across task coordination, vision-guided manipulation, and failure handling, reducing operational failures by 94% and increasing readiness for scalable production deployment.
- Supported the company's first U.S. commercial deployment through live system validation, technical problem-solving, and system stabilization, contributing to customer adoption and subsequent multi-million-dollar expansion.
- Building LLM-agent tooling for automation operations: safety-gated agent skills, diagnostics and validation workflows, and knowledge pipelines with persistent engineering context.

Axle Informatics, Rockville, MD

Automation Research Associate, *Internship*

September 2023 – December 2023

- Owned the design and deployment of a High-Density Storage (HDS) subsystem, enabling reliable integration with Omron mobile robots and PF400 robot arms and contributing to a 37% reduction in lab process time.
- Improved the HDS perception reliability from 53% to 98% through image-preprocessing and shape-detection improvements in OpenCV and Python.

Laboratory for Computational Sensing and Robotics, Baltimore, MD

Research Assistant, *Advisor: Russell H. Taylor*

May 2023 – September 2023

- Optimized malaria vaccine production system setup time by 62% using deep learning-based computer vision.
- Improved ROI detection accuracy to 96% using YOLOv5 and PyTorch-based class activation mapping, achieving 95% vaccine quality prediction accuracy.

Dynacolor, Taipei, Taiwan

R&D Engineer

May 2019 – March 2021

- Engineered mechanical components for a 6-DOF surgical camera system meeting IP66/67 and IK10 requirements.
- Reduced defects by over 80% through SOP development, root-cause analysis, and process optimization.

PROJECTS

Sensor-based Robot Arm Manipulation, Johns Hopkins University

September 2022 – December 2022

- Improved UR5 manipulation performance by reducing control time by over 50% and increasing RRT planning efficiency by 73% through control tuning in MATLAB and ROS and k-d tree implementation in C++.
- Implemented hand-eye calibration using the Park-and-Martin method, achieving 0.2 mm end-effector error.

EDUCATION

Johns Hopkins University, Baltimore, MD

Master of Science in Engineering in Robotics (GPA: 3.9)

August 2022 – May 2024

National Taiwan University, Taipei, Taiwan

Bachelor of Science in Mechanical Engineering

September 2014 – June 2018